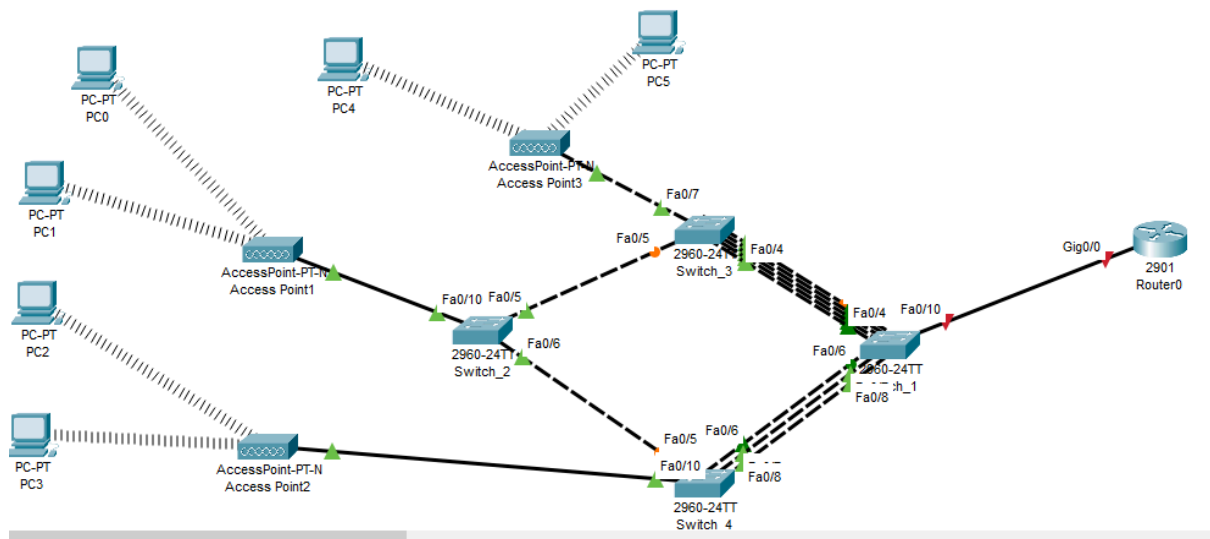


READ_ME

TOPOLOGY



NETWORK ELEMENT

SWITCHES

PCs

AP

Expectation on this configuration

- RSTP
- Root Guard
- BPDU Guard
- Loop prevention
- EtherChannel
- PortFast
- Root Bridge control
- Secondary Root
- Trunking
- VLAN segmentation
- Enterprise failover

NO ROOTGUARD CONFIGURATION WAS IMPLEMENTED BECAUSE ALL THE SWITCHES CONNECTED ARE TRUSTED

SWITCHES

SWITCH_1 (PRIMARY)

Switch(config)#DO SH SPANNING-TREE

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0007.EC8A.3BB0

Cost 6

Port 27(Port-channel1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 000D.BD81.D2A3

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg BLK 19 128.1 P2p

Fa0/3 Desg BLK 19 128.3 P2p

Fa0/6 Desg BLK 19 128.6 P2p

Fa0/4 Desg BLK 19 128.4 P2p

Fa0/5 Altn BLK 19 128.5 P2p

Fa0/2 Desg BLK 19 128.2 P2p

Po1 Root FWD 6 128.27 Shr

VLAN0010

Spanning tree enabled protocol rstp

Root ID Priority 24586

Address 000D.BD81.D2A3

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24586 (priority 24576 sys-id-ext 10)

Address 000D.BD81.D2A3

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p

Fa0/3 Desg FWD 19 128.3 P2p

Fa0/4 Desg FWD 19 128.4 P2p

Fa0/2 Desg FWD 19 128.2 P2p

Po1 Desg FWD 6 128.27 Shr

VLAN0020

Spanning tree enabled protocol rstp

Root ID Priority 24596

Address 000D.BD81.D2A3

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24596 (priority 24576 sys-id-ext 20)

Address 000D.BD81.D2A3

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p
Fa0/3 Desg FWD 19 128.3 P2p
Fa0/4 Desg FWD 19 128.4 P2p
Fa0/2 Desg FWD 19 128.2 P2p
Po1 Desg FWD 6 128.27 Shr
VLAN0030

Spanning tree enabled protocol rstp

Root ID Priority 24606

Address 000D.BD81.D2A3

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24606 (priority 24576 sys-id-ext 30)

Address 000D.BD81.D2A3

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p
Fa0/3 Desg FWD 19 128.3 P2p
Fa0/4 Desg FWD 19 128.4 P2p
Fa0/2 Desg FWD 19 128.2 P2p
Po1 Desg FWD 6 128.27 Shr
VLAN0040

Spanning tree enabled protocol rstp

Root ID Priority 24616

Address 000D.BD81.D2A3

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24616 (priority 24576 sys-id-ext 40)

Address 000D.BD81.D2A3

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Po1 Desg FWD 6 128.27 Shr

spanning-tree mode rapid-pvst

spanning-tree extend system-id

spanning-tree vlan 10,20,30,40 priority 24576

!

interface Port-channel1

switchport trunk allowed vlan 10,20,30,40

switchport mode trunk

!

```

interface FastEthernet0/1
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/4
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active

```

SWITCH_2 (SECORDARY)

```

Switch(config-if)#DO SH SPANNING-TREE
VLAN0001
Spanning tree enabled protocol rstp
Root ID Priority 32769
Address 0007.EC8A.3BB0
Cost 19
Port 5(FastEthernet0/5)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000C.CF62.6992
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/5 Root FWD 19 128.5 P2p
Fa0/6 Desg FWD 19 128.6 P2p
Fa0/10 Desg FWD 19 128.10 Shr

spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree extend system-id
spanning-tree vlan 10,20,30,40 priority 28672
!
interface FastEthernet0/1
!

```

```

interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
spanning-tree portfast
spanning-tree bpduguard enable

```

SWITCH_3

```

Switch(config)#DO SH SPANNING-TREE
VLAN0001
Spanning tree enabled protocol rstp
Root ID Priority 32769
Address 0007.EC8A.3BB0
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0007.EC8A.3BB0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/4 Desg FWD 19 128.4 P2p
Fa0/6 Desg FWD 19 128.6 P2p
Fa0/1 Desg FWD 19 128.1 P2p
Fa0/3 Desg FWD 19 128.3 P2p
Fa0/5 Altn FWD 19 128.5 P2p
Fa0/7 Desg FWD 19 128.7 Shr
Po1 Desg FWD 8 128.27 Shr
VLAN0010
Spanning tree enabled protocol rstp
Root ID Priority 24586
Address 000D.BD81.D2A3

```


Cost 8
Port 27(Port-channel1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)
Address 0007.EC8A.3BB0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type

Fa0/2 Desg BLK 19 128.2 P2p
Fa0/4 Desg BLK 19 128.4 P2p
Fa0/1 Desg BLK 19 128.1 P2p
Fa0/3 Desg BLK 19 128.3 P2p
Po1 Altn BLK 8 128.27 Shr

VLAN0020
Spanning tree enabled protocol rstp
Root ID Priority 24596
Address 000D.BD81.D2A3

Cost 8
Port 27(Port-channel1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
Address 0007.EC8A.3BB0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type

Fa0/2 Desg BLK 19 128.2 P2p
Fa0/4 Desg BLK 19 128.4 P2p
Fa0/1 Desg BLK 19 128.1 P2p
Fa0/3 Desg BLK 19 128.3 P2p
Po1 Altn BLK 8 128.27 Shr

VLAN0030
Spanning tree enabled protocol rstp
Root ID Priority 24606
Address 000D.BD81.D2A3

Cost 8
Port 27(Port-channel1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)
Address 0007.EC8A.3BB0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type

Fa0/2 Desg BLK 19 128.2 P2p
Fa0/4 Desg BLK 19 128.4 P2p
Fa0/1 Desg BLK 19 128.1 P2p

Fa0/3 Desg BLK 19 128.3 P2p
Po1 Altn BLK 8 128.27 Shr
VLAN0040
Spanning tree enabled protocol rstp
Root ID Priority 24616
Address 000D.BD81.D2A3
Cost 8
Port 27(Port-channel1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32808 (priority 32768 sys-id-ext 40)
Address 0007.EC8A.3BB0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type

Po1 Root FWD 8 128.27 Shr

```
spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree extend system-id
!
interface Port-channel1
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/4
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/5
```

```
interface FastEthernet0/6
!  
interface FastEthernet0/7  
spanning-tree portfast  
spanning-tree bpduguard enable
```

SWITCH 4

```
Switch(config)#DO SH SPANNING-TREE  
VLAN0001  
Spanning tree enabled protocol rstp  
Root ID Priority 32769  
Address 0007.EC8A.3BB0  
Cost 25  
Port 6(FastEthernet0/6)  
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec  
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)  
Address 0030.A386.1935  
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec  
Aging Time 20  
Interface Role Sts Cost Prio.Nbr Type  
-----  
Fa0/7 Desg FWD 19 128.7 P2p  
Fa0/6 Root FWD 19 128.6 P2p  
Fa0/5 Altn BLK 19 128.5 P2p  
Fa0/8 Desg FWD 19 128.8 P2p  
Fa0/10 Desg FWD 19 128.10 Shr  
VLAN0010  
Spanning tree enabled protocol rstp  
Root ID Priority 32778  
Address 0030.A386.1935  
This bridge is the root  
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec  
Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)  
Address 0030.A386.1935  
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec  
Aging Time 20  
Interface Role Sts Cost Prio.Nbr Type  
-----  
Fa0/7 Desg FWD 19 128.7 P2p  
Fa0/8 Desg FWD 19 128.8 P2p  
VLAN0020  
Spanning tree enabled protocol rstp  
Root ID Priority 32788  
Address 0030.A386.1935  
This bridge is the root  
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
Address 0030.A386.1935
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type

Fa0/7 Desg FWD 19 128.7 P2p

Fa0/8 Desg FWD 19 128.8 P2p

VLAN0030

Spanning tree enabled protocol rstp

Root ID Priority 32798

Address 0030.A386.1935

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)

Address 0030.A386.1935

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/7 Desg FWD 19 128.7 P2p

Fa0/8 Desg FWD 19 128.8 P2p

spanning-tree mode rapid-pvst

spanning-tree extend system-id

!

interface Port-channel1

switchport trunk allowed vlan 10,20,30,40

switchport mode trunk

!

interface FastEthernet0/1

!

interface FastEthernet0/2

!

interface FastEthernet0/3

!

interface FastEthernet0/4

!

interface FastEthernet0/5

!

interface FastEthernet0/6

!

interface FastEthernet0/7

switchport trunk allowed vlan 10,20,30,40

switchport mode trunk

channel-group 1 mode active

!

```
interface FastEthernet0/8
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/9
!
interface FastEthernet0/10
spanning-tree portfast
spanning-tree bpduguard enable
!
```